

Q 1.1 –

All column data type in customers table –

Field name	Type	Mode	Key	Collation	Default Value	Policy Tags	Description
customer_id	STRING	NULLABLE	-	-	-	-	-
customer_unique_id	STRING	NULLABLE	-	-	-	-	-
customer_zip_code_prefix	INTEGER	NULLABLE	-	-	-	-	-
customer_city	STRING	NULLABLE	-	-	-	-	-
customer_state	STRING	NULLABLE	-	-	-	-	-

Inference:

- Data type of column customer_id, customer_unique_id, customer_city, customer_state is STRING.
- Data type of column customer_zip_code_prefix is INTEGER.

Q 1.2

Query:

```
3  
4 select min(order_purchase_timestamp) as first_order_date,  
5        max(order_purchase_timestamp) as last_order_date  
6 from Target.orders
```

Query output:

Row	first_order_date	last_order_date
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

Inference:

- First order was placed on 4th September 2016 and last order was placed on 17th October 2018.
- Orders table contains order details of total 26 months from 2016 to 2018.

Q 1.3

Query:

```
8 select count(distinct customer_city) as no_of_city,  
9      count(distinct customer_state) as no_of_state  
10 from `Target.customers`
```

Query output:

Query results			
Press Alt+T for Accessibility Option			
SAVE RESULTS EXPLORE DATA			
JOB INFORMATION RESULTS CHART JSON EXECUTION DETAILS EXECUTION GRAPH			
Row	no_of_city	no_of_state	
1	4119	27	

Inference:

- From 4th September 2016 to 10th October 2018 customers from 4119 different cities and 27 states ordered from target.
- Target is getting orders from all the 27 states of Brazil.

Q 2.1

Query:

```
11  
12 with table1 as  
13 (select *,  
14      extract(year from order_purchase_timestamp) as order_year  
15 from `Target.orders`)  
16  
17 select order_year,  
18      count(distinct order_id)  
19 from table1  
20 group by 1  
21 order by 1
```

Query output:

Query results			
JOB INFORMATION RESULTS CHART			
Row	order_year	no_of_orders	
1	2016	329	
2	2017	45101	
3	2018	54011	

Inference:

- In year 2018 and 2017 no. of orders placed has increased significantly as compared to orders placed in year 2016.
- There is increase in number of orders placed in 2018 compared to no of orders placed in 2017.
- So no. of orders placed are increasing year by year therefore company should anticipate more no. of orders in coming years and manage stock of products accordingly so that there will not be shortage of products.

Q 2.2

Query:

```
with table1 as
(select *,
    format_date('%Y-%m', order_purchase_timestamp) as order_month_year
from `Target.orders`)

select order_month_year,
    count(distinct order_id) as no_of_orders
from table1
group by 1
order by 1
```

Query output:

Query results

 SAVE RESULTS ▾

 EXPLORE DATA ▾

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	order_month_year ▾	no_of_orders ▾				
1	2016-09	4				
2	2016-10	324				
3	2016-12	1				
4	2017-01	800				
5	2017-02	1780				
6	2017-03	2682				
7	2017-04	2404				
8	2017-05	3700				
9	2017-06	3245				
10	2017-07	4026				

Inference:

- In year 2016 highest no of orders were placed in October month and least orders were placed in December month.
- In year 2017, in month OCT NOV DEC highest number of orders were placed and least no. of orders were placed in first quarter of year.
- In year 2018 highest number of orders were placed in January February and March month and least orders were placed in last quarter of the year.
- Number of orders placed are increasing towards the end of the year and year by year.

Q 2.3

Query:

```
with table1 as
(select *,
    format_date('%T', order_purchase_timestamp) as order_time
from `Target.orders`),

table2 as
(select *,
    case when order_time between '00:00:00' and '05:59:59' then 'Dawn'
    when order_time between '06:00:00' and '11:59:59' then 'Morning'
    when order_time between '12:00:00' and '17:59:59' then 'Afternoon'
    when order_time between '18:00:00' and '23:59:59' then 'Night'
    end as time_of_day
from table1)

select time_of_day,
    count(distinct order_id) as no_of_orders
from table2
group by 1
order by no_of_orders desc
```

Query output:

Query results

JOB INFORMATION		RESULTS	CHART	JSON
Row	time_of_day	no_of_orders		
1	Afternoon	38361		
2	Night	34100		
3	Morning	22240		
4	Dawn	4740		

Inference:

- Customers are placing orders mostly during afternoon, night & morning.
- Customers are placing least number of orders during dawn.

Q 3.1

Query:

```
with table1 as
(select *,
    format_date('%Y-%m', order_purchase_timestamp) as order_month_year
from `Target.orders`)

select b.customer_state,
    a.order_month_year,
    count(distinct order_id) as no_of_order
from table1 a
left join `Target.customers` b
on a.customer_id = b.customer_id
group by 1,2
order by b.customer_state asc, a.order_month_year
```

Query output:

Query results				SAVE RESULTS	EXPLORE DATA	↺
JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	customer_state	order_month_year	no_of_order			
1	AC	2017-01	2			
2	AC	2017-02	3			
3	AC	2017-03	2			
4	AC	2017-04	5			
5	AC	2017-05	8			
6	AC	2017-06	4			
7	AC	2017-07	5			
8	AC	2017-08	4			
9	AC	2017-09	5			
10	AC	2017-10	6			

Inference:

- Sao Paulo has highest number of orders paced in single month. So next year we should anticipate more number of orders from this state in this month and ensure that there will be proper and adequate supply of products.
- Minas Gerais and Rio De Janero has orders placed between 900 to 1100 in single month.
- Rio Grande do Sul has third highest orders placed in single month.

Q 3.2

Query:

```
select customer_state,
    count(distinct customer_unique_id) as no_of_customers
from `Target.customers`
group by 1
order by 2 desc
```

Query output:

Row	customer_state	no_of_customers
1	SP	40302
2	RJ	12384
3	MG	11259
4	RS	5277
5	PR	4882
6	SC	3534
7	BA	3277
8	DF	2075
9	ES	1964
10	GO	1952

Inference:

- Sao Paulo has highest number of customers ordering from Target.
- Minas Gerais and Rio De Janero has second highest customers.
- Rio Grande do Sul and Parana has third highest number of customers.
- We should ensure that there will not be any disruptions in supply chain in the states with highest customers and more number of orders.

Q 4.1

Query:

```
with table1 as
(select a.order_id, a.customer_id,a.order_purchase_timestamp, b.payment_value,
      format_date('%Y-%m', a.order_purchase_timestamp) as order_month_year
from `Target.orders` a
left join `Target.payments` b
on a.order_id = b.order_id),
table2 as
(select 1 as rn,
      round(sum(payment_value),2) as total_payment_in_2017
from table1
where order_month_year like '2017%' and order_month_year < '2017-09'),

table3 as
(select 1 as rn,
      round(sum(payment_value),2) as total_payment_in_2018
from table1
where order_month_year like '2018%' and order_month_year < '2018-09')

select a.total_payment_in_2017,b.total_payment_in_2018,
      round(((b.total_payment_in_2018 -
a.total_payment_in_2017)*100)/a.total_payment_in_2017,2) as pct_increase_from_2017_to_2018
from table2 a
left join table3 b
```

Query output:

Query results

JOB INFORMATION

RESULTS

CHART

JSON

EXECUTION DETAILS

EXECUTION

Row	total_payment_in_2017	total_payment_in_2018	pct_increase_from_2017_to_2018
1	3669022.12	8694733.84	136.98

Inference:

- From 2017 to 2018 there is 136 % increase in total payment for the months January to August.
- Company can anticipate more number of orders in coming years as it is evident from the above data that total order value has increased significantly from 2017 to 2018.
- We should look for more sellers and delivery executives in the states where demand is high and arrange stock of product in advance.

Q 4.2

Query:

```
select distinct b.seller_state,
               round (avg(a.price) over (partition by b.seller_state),2) as avg_price,
               round(sum(a.price) over (partition by b.seller_state),2) as total_price
from `Target.order_items` a
left join `Target.sellers` b
on a.seller_id = b.seller_id
order by 2 desc, 3 desc
```

Query output:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS
Row	seller_state	avg_price	total_price		
1	PB	449.87	17095.0		
2	BA	444.11	285561.56		
3	AM	392.33	1177.0		
4	RO	340.16	4762.2		
5	AC	267.0	267.0		
6	CE	215.33	20240.64		
7	PI	210.17	2522.0		
8	PE	204.23	91493.85		
9	RN	178.44	9992.6		
10	RJ	175.17	843984.22		

Inference:

- Paraiba state has highest average product price followed by state Bahia and Amazonas, long freight distance and less sellers might be the reason for more product cost in these states.
- Sao Paulo and Parana state has highest total price as we are getting maximum number of orders from these states.

Q 4.3

Query:

```
select distinct b.seller_state,
               round (avg(a.freight_value) over (partition by b.seller_state),2) as
avg_freight_value,
               round(sum(a.freight_value) over (partition by b.seller_state),2) as
total_freight_value
from `Target.order_items` a
left join `Target.sellers` b
on a.seller_id = b.seller_id
order by 2 desc, 3 desc
```


Query output:

Row	seller_state	avg_freight_value	total_freight_value
1	RO	50.91	712.78
2	CE	46.38	4359.83
3	PB	39.19	1489.15
4	PI	36.94	443.32
5	AC	32.84	32.84
6	ES	32.72	12171.13
7	MT	31.94	4631.73
8	SE	31.85	318.49
9	BA	30.64	19700.68
10	MA	29.98	12141.29

Inference:

- State Rondonia and Ceara have highest freight value.
- Company should try to find out the reasons why freight value in these states is high in order to reduce cost of freight in these states.
- We can analyse distribution of sellers across these states and look for more sellers to increase the density of sellers in these states, this will reduce the time and cost of freight and also total amount paid by customer on each order/product.

Q 5.1

Query:

```
select customer_id,  
       date_diff( order_delivered_customer_date,order_purchase_timestamp,day) as  
time_to_deliver,  
       date_diff(order_delivered_customer_date , order_estimated_delivery_date,day) as  
diff_estimated_delivery  
from `Target.orders`  
order by 2 desc, 3 desc
```

Query output:

Query results

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS
Row	customer_id	time_to_deliver	diff_estimated_delive		
1	75683a92331068e2d281b11a...	209	181		
2	d306426abe5fca15e54b645e4...	208	188		
3	7815125148cfa1e8c7fee1ff79...	195	165		
4	9cf2c3fa2632cee748e1a59ca9...	194	166		
5	1a8a4a30dc296976717f44e78...	194	161		
6	217906bc11a32c1e470eb7e08...	194	155		
7	cb2caaaead400c97350c37a3f...	191	175		
8	65b14237885b3972ebec28c0f...	189	167		
9	8199345f57c6d1cbe9701f924...	188	159		
10	f85e9ec0719b16dc4dd0edd43...	187	162		

Inference:

- Maximum time to deliver and estimated time to deliver are 209 & 181 days respectively.
- Time to deliver varies from minimum 0 day to maximum of 209 days.

Q 5.2

Top five state with highest freight value

Query:

```
select distinct b.seller_state,
               round (avg(a.freight_value) over (partition by b.seller_state),2) as
avg_freight_value

from `Target.order_items` a
left join `Target.sellers` b
on a.seller_id = b.seller_id
order by 2 DESC
limit 5
```

Query output:

Query results

JOB INFORMATION		RESULTS	CHART	JSON
Row	seller_state	avg_freight_value		
1	RO	50.91		
2	CE	46.38		
3	PB	39.19		
4	PI	36.94		
5	AC	32.84		

Top five state with least freight value

Query:

```
select distinct b.seller_state,  
               round (avg(a.freight_value) over (partition by b.seller_state),2) as  
               avg_freight_value  
  
from `Target.order_items` a  
left join `Target.sellers` b  
on a.seller_id = b.seller_id  
order by 2 asc  
limit 5
```

Query output:

Query results			
JOB INFORMATION		RESULTS	CHART
Row	seller_state	avg_freight_value	
1	SP	18.45	
2	PA	19.39	
3	RJ	19.47	
4	DF	20.57	
5	PR	22.72	

Inference:

- Sao Paulo state has minimum average freight value followed by para and Rio de Janero.
- State Rondonia and Ceara have highest freight value, so cost of freight movement is high in these states.
- In the states where average freight value is more company should look for more sellers in nearby places from where we are getting orders so that freight value will reduce which will reduce overall product cost also.

Q 5.3

Top five state with highest average delivery time

Query:

```
with table1 as
(select a.order_id,b.customer_state,
       date_diff( a.order_delivered_customer_date,a.order_purchase_timestamp,day) as
time_to_deliver
from `Target.orders` a
left join `Target.customers` b
on a.customer_id = b.customer_id)

select distinct customer_state, avg(time_to_deliver) over (partition by customer_state)
avg_delivery_time
from table1
order by avg_delivery_time desc
limit 5
```

Query output:

Query results			
JOB INFORMATION		RESULTS	CHART
Row	customer_state	avg_delivery_time	
1	RR	28.97560975609...	
2	AP	26.73134328358...	
3	AM	25.98620689655...	
4	AL	24.04030226700...	
5	PA	23.31606765327...	

Top five state with lowest average delivery time –

Query:

```
with table1 as
(select a.order_id,b.customer_state,
       date_diff( a.order_delivered_customer_date,a.order_purchase_timestamp,day) as
time_to_deliver
from `Target.orders` a
left join `Target.customers` b
on a.customer_id = b.customer_id)

select distinct customer_state, avg(time_to_deliver) over (partition by customer_state)
avg_delivery_time
from table1
order by avg_delivery_time asc
limit 5
```

Query output:

Query results			
JOB INFORMATION		RESULTS	CHART
Row	customer_state	avg_delivery_time	
1	SP	8.298061489072...	
2	PR	11.52671135486...	
3	MG	11.54381329810...	
4	DF	12.50913461538...	
5	SC	14.47956019171...	

Inference:

- Sao Paulo and para has lowest delivery time whereas Roraima and Amapa has highest delivery time.
- Company should look for more sellers in nearby areas from where company is receiving orders to reduce the delivery time.
- Consider other options for freight movement in the states where delivery time is higher for reducing the delivery time.

Q 5.4

Top five states with fastest delivery

Query:

```
with table1 as
(select a.order_id,b.customer_state,
       date_diff( a.order_delivered_customer_date,a.order_purchase_timestamp,day) as
actual_time_to_deliver,
       date_diff(a.order_estimated_delivery_date,a.order_purchase_timestamp,day) as
estimated_time_to_deliver
from `Target.orders` a
left join `Target.customers` b
on a.customer_id = b.customer_id),

table2 as
(select distinct customer_state,
       avg(actual_time_to_deliver) over (partition by customer_state) as
avg_actual_time_to_deliver,
       avg(estimated_time_to_deliver) over(partition by customer_state) as
avg_estimated_time_to_deliver
from table1)

select customer_state,
       (avg_estimated_time_to_deliver - avg_actual_time_to_deliver) as no_of_days
from table2
order by 2 desc
limit 5
```

Query output:

Query results			
JOB INFORMATION		RESULTS	CHART
Row	customer_state	no_of_days	
1	AC	20.12793209876...	
2	RO	19.49353437759...	
3	AP	18.97453906935...	
4	AM	18.77054986020...	
5	RR	17.19830328738...	

Inference:

- Acre, Rondonia and Amapa states are having fastest delivery time as compared to the estimated delivery time.
- Based on the customer, seller and delivery executive feedbacks we can analysed why delivery in these states is faster compared to other states. Based on the findings we can apply same in other states so that the delivery time will improve in other states also wherever feasible.
- Based on the findings we can apply same in other states so that the delivery time will improve in other states also wherever feasible.

Q 6.1

Query:

```
with table1 as
(select *,
    format_date('%Y-%m', order_purchase_timestamp) as order_month_year
from `Target.orders`)

select b.payment_type,
    a.order_month_year,
    count(distinct a.order_id) as no_of_order
from table1 a
left join `Target.payments` b
on a.order_id = b.order_id
group by 1,2
order by 2
```

Query Output :

Query results

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS
Row	payment_type	order_month_year	no_of_order		
1	credit_card	2016-09	3		
2	null	2016-09	1		
3	credit_card	2016-10	253		
4	UPI	2016-10	63		
5	voucher	2016-10	11		
6	debit_card	2016-10	2		
7	credit_card	2016-12	1		
8	credit_card	2017-01	582		
9	UPI	2017-01	197		
10	voucher	2017-01	33		

Inference :

- Maximum number of customers are using credit card for making payment.
- Least number of customers are using debit card for making payment.
- UPI is second mostly used mode of payment for making payment.

Q 6.2

Query:

```
select payment_installments,
       count(distinct order_id) as no_of_orders
from `Target.payments` b
where b.payment_installments >=1
group by 1
order by 2 desc
```

Query output:

Query results			
JOB INFORMATION		RESULTS	CHART
Row	payment_installment	no_of_orders	
1	1	49060	
2	2	12389	
3	3	10443	
4	4	7088	
5	10	5315	
6	5	5234	
7	8	4253	
8	6	3916	
9	7	1623	
10	9	644	

Inference:

- Customers who have made payment with single instalment are highest.
- Customers who have made payment in two instalment are second highest.
- Maximum number of customers are making payment in one instalment only.